Entomological Investigations in Chichaoua: An Emerging Epidemic Focus of Cutaneous Leishmaniasis in Morocco

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ABSTRACT Cutaneous leishmaniasis due to *Leishmania tropica* Wright seems to be an emerging disease in Chichaoua, a province located in southwestern Morocco. In this study, sand flies (Diptera: Psychodidae) were collected from 12 stations. Sticky traps were placed in domestic, peridomestic, and sylvatic sites. In total, 3,787 specimens consisting of 10 species (seven *Phlebotomus* and three *Sergentomiya*) were identified. *Phlebotomus perniciosus* Newstead, the predominant species, was abundant, especially in mountainous areas. *Phlebotomus sergenti* Parrot (12%) was found in all studied villages where it was associated with domestic and peridomestic habitats. On the basis of its abundance, distribution, and notable anthropophily, *P. sergenti*, a proven vector of *L. tropica* elsewhere, is considered the cutaneous leishmaniasis vector in this emerging focus.

KEY WORDS cutaneous leishmaniasis, sand fly, *Phlebotomus sergenti*, *Leishmania tropica*, Morocco

IN MOROCCO, CUTANEOUS LEISHMANIASIS (CL) is caused specifically by *Leishmania major* Yakimoff and Schokhor and *Leishmania tropica* Wright (Rioux et al. 1986b, Pratlong et al. 1991). Infections due to a dermotropic variant of *Leishmania infantum* Nicolle also have been found (Rioux et al. 1996, Lemrani et al. 1999). CL caused by *L. tropica* is widespread in the central and southern semiarid zones Morocco. After the first case was reported in the High-Atlas mountains (Marty et al. 1989), four foci were identified in Tanant, Smimou, Taza, and Zouagha Moulay Yaacoub (Guilvard et al. 1991, Pratlong et al. 1991, Guessous Idrissi et al. 1997, Rhajaoui et al. 2004). In the past 4 yr, the reported cases of CL due to *L. tropica* have increased in Morocco (Figs. 1 and 2), and new foci have been identified in some nonendemic areas, mainly in Zouagha Moulay Yaacoub (Rhajaoui et al. 2004) and Chichaoua. According to the Moroccan Ministry of Public Health (MPH 2001), CL due to *L. tropica* is an emerging disease in the province of Chichaoua where 1,025 cases were officially reported between 2000 and 2002.

This article presents the results of an entomological survey carried out in a CL focus in Chichaoua Province, Morocco, with the aim of describing the species composition of the sand fly fauna, their ecology, and monthly prevalence and of identifying the probable vector of CL.

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Materials and Methods

Study Area. Field studies were conducted in 12 stations in Chichaoua Province (Fig. 3), Morocco. The climate is arid to semiarid, with monthly mean high and low temperatures ranging from 38.3 to 5.4°C in July and January, respectively. The average annual rainfall is 270 mm.

The province covers an area of 7,120 km^2 with altitude ranging from 322 to 1,446 m above sea level. It can be divided into two parts. In the northern part (Imi'Ntanout, Tachmirou, Sidi Ghanem, Sidi Abdelmoumen, Tadnest, Bouabout, Chichaoua, and Mejjat), the vegetation is composed mainly of jujube, *Zizyphus lotus* Lamarck, trees characteristic of arid areas. In the southern part (Timsal, Aït Moussa, Lalla Aziza, and Aït Mhaned), the topography is rugged and the vegetation is more diversified (olive, almond, and carob trees) and characteristic of that found in semiarid areas. The total population is 343,068 inhabitants, composed mainly of farmers.

Sand Fly Collection and Identification. Sand flies were collected in July 2001 and from July 2002 to December 2003. Collections were carried out twice monthly between July and November, and monthly between December and June. Sticky traps, sheets of papers (20 by 20 cm) coated with castor oil, were placed in intradomestic, peridomestic, and sylvatic sites. Peridomestic sites included areas adjacent to human dwellings and contained housing for domestic animals (dogs, chickens, and cows). Sylvatic sites were at least a 500-m distance from human dwellings. In these sites, traps were placed close to vegetation, river edges, crevices, and caves. Traps were set in the af-

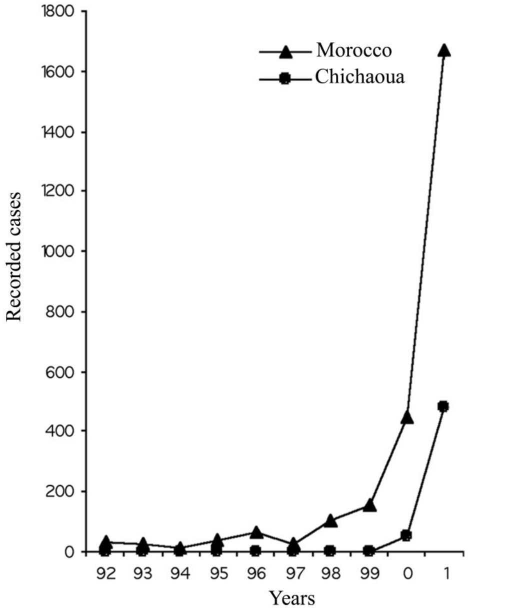

Fig. 1. Number of cases of *L. tropica* reported annually in Chichaoua and throughout Morocco from 1992 to 2001 (MPH 2001).

ternoon and collected the next morning. Data are presented as the number of sand flies per square meter of sticky trap.

Collected sand flies were stored in 70% ethanol. Species identification was made by examining the morphology of the pharyngeal armatures and spermathecae of female flies and the external genitalia of males, by using the keys and descriptions presented by Léger et al. (1983), MPH (1997), Benabdennbi et al. (1999), and Depaquit et al. (2000).

Results

During the 3-yr study (2001–2003), 3,787 sand flies were collected. They comprised 10 phlebotomine species with seven *Phlebotomus* spp. and three *Sergentomyia* spp. (Table 1). Among the *Phlebotomus* spp., *Phlebotomus perniciosus* Newstead was the predominant species (45.8%) followed by *Phlebotomus sergenti* Parrot (12%), *Phlebotomus longicuspis* Nitzulescu (8.5%), *Phlebotomus ariasi* Tonnoir (5.6%), *Phlebotomus papatasi* Scopoli (4.2%), *Phlebotomus alexandri* Sinton (1.3%), and *Phlebotomus mariae* Rioux et al. (0.1%).

For the genus *Sergentomiya, Sergentomiya fallax* Parrot was the most prevalent species (15.1%). *Sergentomiya minuta* Rondani and *Sergentomiya dreyfussi* Parrot accounted for 4.3 and 3.2%, respectively.

Fluctuations in abundance of sand flies were observed from July 2002 to December 2003. During this period, sand flies were active each year from June through late November (Fig. 4). Peak population densities for each species occurred as follows: *P. sergenti,* July–August; *P. longicuspis,* August–September; *P. perniciosus,* September–October; and *P. ariasi,* late October.

The prevalence of *Phlebotomus* species was compared in domestic, peridomestic, and sylvatic microhabitats in the arid areas of Imi'Ntanout, Tachmirou, and Sidi Abdelmoumen (Fig. 5), and in the semiarid areas of Lalla Aziza, Timsal, and Aït Moussa (Fig. 6).

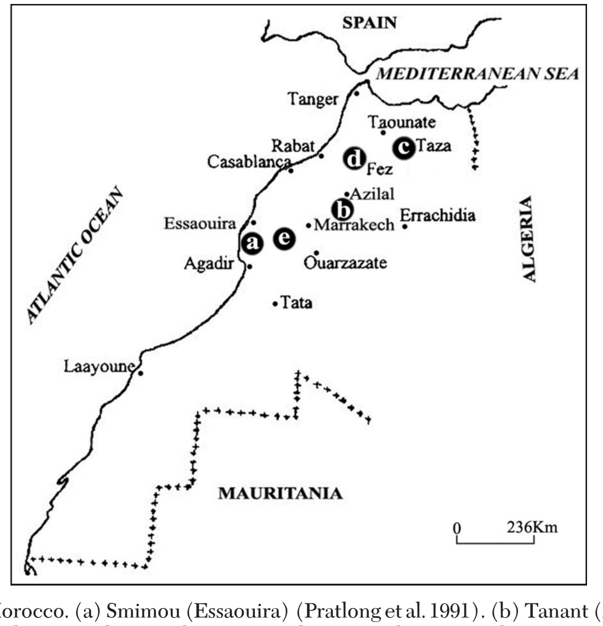

Fig. 2. *L. tropica* foci in Morocco. (a) Smimou (Essaouira) (Pratlong et al. 1991). (b) Tanant (Azilal) (Pratlong et al. 1991). (c) Taza (Guessous Idrissi et al. 1997). (d) Zouagha My Yacoub (Fes) (Rhajaoui et al. 2004). (e) Chichaoua (current study).

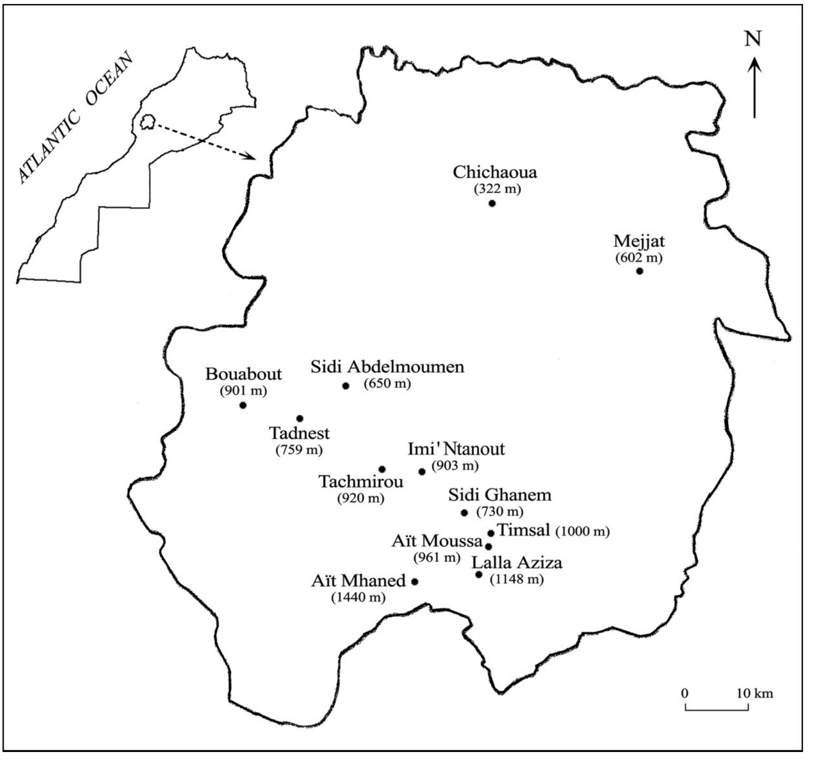

Fig. 3. Map of Chichaoua Province showing the main study villages where sand flies were traps; the altitude above sea level of each village is given in parentheses.

Overall, collections from semiarid biotopes showed a higher mean density and were richer in their species composition compared with arid biotopes (collections included *P. ariasi* and *P. alexandri*). *P. perniciosus* was the most abundant species in semiarid areas but was replaced by *P. longicuspis* in arid areas. *P. sergenti* was the second most abundant species in both areas but was considerably more abundant in semiarid habitats. Most of the *Phlebotomus* species were collected in domestic and peridomestic microhabitats (Figs. 5 and 6). *P. sergenti,* a highly anthropophilic species, was collected primarily inside and around the houses.

Table 1. Sand fly species collected in Chichaoua Province and their relative abundance (%)

Subgenus	Species	Male	Female	Total	%
Phlebotomus	*P. papatasi*	114	44	158	4.2
Paraphlebotomus	*P. sergenti*	330	120	450	12.0
	P. alexandri	23	27	50	1.3
Larroussius	*P. perniciosus*	1421	314	1735	45.8
	P. longicuspis	226	96	322	8.5
	P. ariasi	137	75	212	5.6
	P. mariae	4	0	4	0.1
Sergentomiya	*S. fallax*	206	366	572	15.1
	S. minuta	63	100	163	4.3
Grassomiya	*S. dreyfussi*	64	57	121	3.2
	Total	2,588	1,199	3,787	100.0

Discussion

In Chichaoua, CL due to *L. tropica* is a serious and increasing public health problem (MPH 2001). This paper presents the first report of sand flies in this focus. Ten species (seven of *Phlebotomus* and three of *Sergentomyia*) were collected and identified. *P. perniciosus,* a proven vector of *L. infantum* in the Mediterranean basin (Killick-Kendrick 1990), is the most abundant species in Chichaoua, especially in mountainous areas.

P. alexandri accounts for 1.3% of sand flies collected and was found only in some villages in the region of Chichaoua. This species has never been implicated in the transmission of CL in Morocco.

P. papatasi, the vector of *L. major* in Morocco (Rioux et al. 1986a), is not susceptible to *L. tropica* (Killick-Kendrick et al. 1994). In Chichaoua, there have been no cases of CL caused by *L. major.*

P. sergenti is a proven vector of *L. tropica* in Morocco and elsewhere (Al Zahrani et al. 1988, Pratlong et al. 1991). In Chichaoua, *P. sergenti* exhibited a large ecological plasticity. It was abundant in the arid area of Imi'Ntanout as well as in the semiarid area of Lalla Aziza. The wide distribution and abundance suggest that this species is able to adapt to different bioclimates but has a marked preference for semiarid habitats (Rioux et al. 1984). In this study, the *P. sergenti* population reached peak density during July and

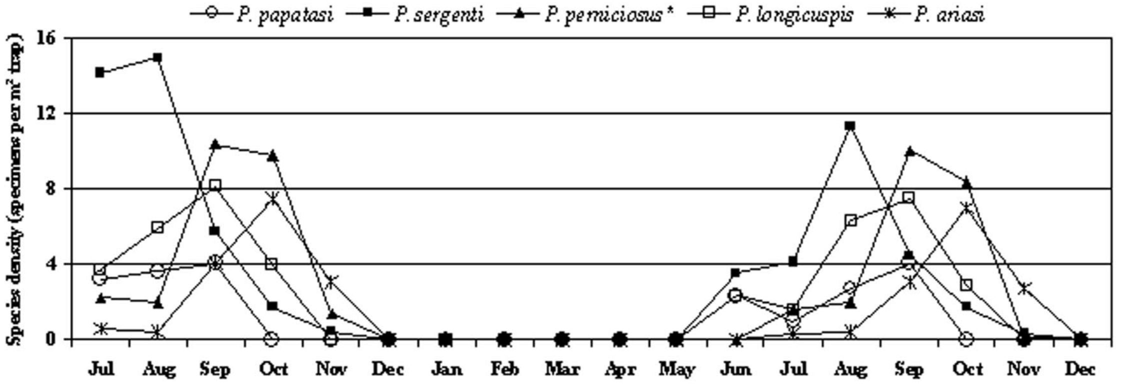

Fig. 4. Seasonality of *Phlebotomus* species in Chichaoua Province monitored from July 2002 to December 2003. *, density multiplied by 10^{-1}.

August and was lowest from September through November. In Marrakech city, *P. sergenti* also was active in July (Boussaa et al. 2004). This anthropophilic species was collected specially in domestic and peridomestic habitats. Additionally, *P. sergenti* is highly susceptible to *L. tropica* (Killick-Kendrick et al. 1995). Based upon these data, *P. sergenti* is considered the most likely vector of CL in Chichaoua.

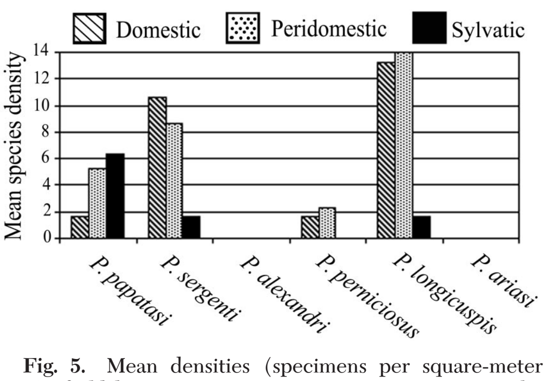

Fig. 5. Mean densities (specimens per square-meter trap) of *Phlebotomus* species in various environments in the arid area of Imintanout, Tachmirou, and Sidi Abdelmoumen.

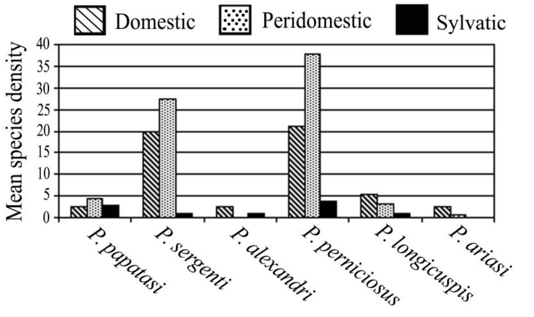

Fig. 6. Mean densities (specimens per square-meter trap) of *Phlebotomus* species in various environments in the semiarid area of Lalla Aziza, Timsal, and Aït Moussa.

However, further investigations are necessary to isolate *Leishmania* strains from *P. sergenti* and to characterize the genotype of Chichaoua sand flies, taking into account current findings about the genetic polymorphism of *P. sergenti* in Morocco (Yahia et al. 2004).

In conclusion, based on this survey, Chichaoua seems to be a typical focus of *P. sergenti* and *L. tropica.* It should be added to the list of CL foci due to *L. tropica,* so we propose a new update map of the distribution of these foci in Morocco (Fig. 2). With *P. sergenti* being domestic and peridomestic, spraying with residual insecticides inside and around houses could be an efficient control method.

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